Recursive Feedback Loops: A Cybernetic–Symbolic Framework for Cognition, Strange Loops, and Human–Feline Interaction

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Abstract

This thesis unifies several frameworks developed by Joseph Louis Riccio into a single cybernetic–symbolic model of *recursive feedback loops*. Drawing on the Thought Tree method (trunk/branch/leaf), formal equations for cognitive state, recursive self-modeling, strange-loop theory, and lived experimental designs, it integrates mathematics, symbolic logic, philosophy, and applied AI research. Recursion is treated as a generative principle across cognition, consciousness, cybernetics, and identity.

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Chapter 1

Introduction

Recursive feedback loops appear wherever systems *sense*, *decide*, and *act* while adapting to the consequences of their own actions. They are the connective tissue across cybernetics, cognitive science, and systems design. In Joseph Riccio's body of work, four strands recur: (i) **analog → digital** translation of behavior into computable symbols; (ii) **dialogue as a loop** (initiation → response → adjustment); (iii) a **state algebra of presence** for humans and felines; and (iv) **staged workflow logic** for process progression.

The thesis assembles these artifacts into one coherent model. We will (1) formalize recursive feedback at multiple scales; (2) specify a pipeline from raw audio/video to symbolic states; (3) present an empirical protocol for measuring feedback; and (4) encode progress with staged logic while keeping the system interpretable.

Overview Figure

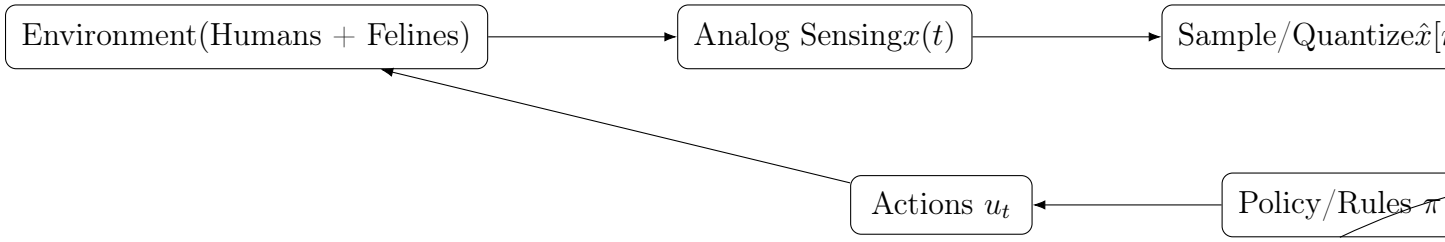


Figure 1.1: Core recursive loop: sensing → symbolizing → state → policy → action, with short- and long-range feedback.

Positioning. The conceptual lineage runs from cybernetics and Hofstadter's *strange loops* to modern recursive AI and Bayesian updating112†source. Your *Thought Tree Method* provides a visual grammar for organizing complex ideas114†source, while your *Theory 2* and *Equation of the Mind* supply formal equations for attention, coupling, and equilibrium111†source113†source.

Chapter 2

Background and Literature

2.1 Strange Loops and Self-Reference

Hofstadter's strange loops describe hierarchical systems that, through self-reference, return to their starting point, producing identity and meaning. This lens clarifies how cognition can model itself and how policies can update themselves from their own outcomes112†source.

2.2 Cybernetics and Feedback

Classic negative feedback stabilizes; positive feedback amplifies novelty. Real cognitive systems blend both, often at different scales (micro sensorimotor vs. macro interpersonal). The recursive composition of loops yields emergent dynamics112†source.

2.3 Quantum Analogies (Exploratory)

Quantum measurement embeds the observer into system evolution, a feedback-like dependency. While speculative for brain mechanisms, the analogy motivates *probabilistic* and *multi-hypothesis* inference in AI112†source.

2.4 Visual Logic for Recursion

Complex loops are difficult to keep in working memory; external visual logic (graphs, staged diagrams) improves reasoning and collaboration112†source. Your Thought Tree contributes a disciplined, recursive visual method114†source.

Chapter 3

The Thought Tree Method

3.1 Trunk, Branches, Leaves

The **trunk** encodes a stable core aim; **branches** are stable support pillars; **leaves** are exploratory ideas that may be pruned or grafted. This provides a living structure for research programs and personal decision-making114†source.

3.2 Workflow

1. Define core idea (trunk).
2. Establish branches (domains or subprograms).
3. Generate leaves (hypotheses/experiments).
4. Integrate or prune leaves based on evidence.

Advantages include clarity, systematic experimentation, and cognitive offloading; risks include overgrowth without pruning and tooling overhead114†source.

3.3 Formal Links to Equations

Your *Theory 2* notes attach symbolic equations to stages and equilibria, enabling the tree to drive math and vice versa111†source.

Chapter 4

Formal Mathematical Framework

4.1 Cognitive Initialization and Transformation

From *Theory 2*111†source:

$$A = \int a_1(a)\, P \;+\; \int a_2(b)\, L,$$
$$A_{31}^{t} = (a_t + a_2) = S_5,$$

with equilibrium conditions

$$e_{1i} + s + r_2 = R_1, \qquad e_{2i} + Jt = R_2.$$

These capture initialization (A), transformation (A_{31}^{t}), and equilibria (R_1, R_2) as recursive cognitive states.

4.2 Global Mind State and Attention

From *Equation of the Mind*113†source:

$$M(t) = \sum_{i=1}^{N} a_i(t)\, y_i(t), \tag{4.1}$$

$$S_i(t) = \alpha\, \mathrm{align}(B_i, V(t)) + \beta\, \mathrm{urgency}_i(t) + \gamma\, \mathrm{novelty}_i(t) + \delta\, \mathrm{reward}_i(t) - \kappa\, \mathrm{cost}_i(C(t)), \tag{4.2}$$

$$a_i(t) = \frac{\exp(S_i(t))}{\sum_{j=1}^{N} \exp(S_j(t))}. \tag{4.3}$$

Branch outputs fuse logic, perception, experience, recursion:

$$y_i(t) = \phi_i\big(W_i^L L_i + W_i^P P_i + W_i^E E_i + W_i^R R_i + b_i\big). \tag{4.4}$$

4.3 Constraints, Coupling, and Special Modules

$$\sum_{i=1}^{N} a_i(t)\,\mathrm{depth}_i(t) \le D_{\max}(t) = h(C(t)), \tag{4.5}$$

$$\tilde{y}_i(t) = y_i(t) + \sum_{j\neq i} \Gamma_{ij} y_j(t-1). \tag{4.6}$$

Selected special modules (illustrative)113†source:

$$T_{\text{Tenchi}}(t{+}1) = \alpha_T T(t) + \beta_T\,\mathrm{pos}(t) - \gamma_T\,\mathrm{neg}(t), \tag{4.7}$$

$$\mathrm{Addiction}(t{+}1) = \lambda\,\mathrm{Addiction}(t) + \mu\,\mathrm{stress}(t) - \nu\,\mathrm{supports}(t). \tag{4.8}$$

4.4 Stage–Task Logic (Symbolic)

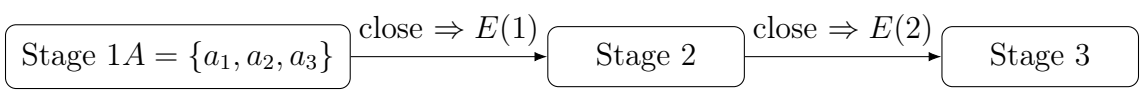
We encode closure, enablement, and effect tokens:

$$\mathrm{Closed}(s) \;\Leftrightarrow\; \forall x\,(x \prec s \Rightarrow \mathrm{Done}(x)), \tag{4.9}$$

$$\mathrm{Closed}(s) \Rightarrow \mathrm{Enabled}(\mathrm{next}(s)), \tag{4.10}$$

$$E(1) = st_1, \quad E(2) = st_2, \quad R_1 = st_1 + st_1, \quad R_2 = st_2 + st_1. \tag{4.11}$$

This mirrors your stage boards (EIN, LLC(MN), certification in Stage 1)111†source.



$$R_1 = st_1 + st_1,\; R_2 = st_2 + st_1$$

Figure 4.1: Stage progression with effect tokens (4.11).

Chapter 5

Strange Loops and Quantum-Inspired Recursion

5.1 Self-Reference and Identity

Strange loops explain how systems re-enter their own descriptions to generate identity. This is relevant to meta-cognition and policy self-updating112†source.

5.2 Quantum-Inspired Update (Conceptual)

A compact recursive update:

$$\Psi(t{+}1) = F(\Psi(t), O(t)), \qquad P(t{+}1) = \left|\Psi(t{+}1)\right|^2, \tag{5.1}$$

treats decisions as probabilistic collapses of internally simulated hypotheses112†source.

Chapter 6

Recursive Self-Modeling

Your *Recursive Self-Modeling Spec* encodes an agent that keeps a model of *itself* inside its world model. Key ingredients:

- A reflective state that tracks its own policies and uncertainty.
- A meta-update that adjusts policies using summaries of prior trajectories.
- Visual logic (graphs, trees) to externalize the recursion and keep it legible112†source.

Chapter 7

Experiment I: Dialogue Feedback (Haru Context)

7.1 Purpose and Focus

Measure feedback during dialogue as knowledge and affect evolve. Focus on information sharing, agreement/disagreement, and understanding shifts.

7.2 Protocol

1. Two participants with differing knowledge baselines.
2. Multimodal capture (audio/video); transcript with timestamps.
3. Annotate *Initiation* $\rightarrow$ *Response* $\rightarrow$ *Adjustment* cycles.

7.3 Metrics

- Response latency distributions and turn-taking asymmetry.
- Transfer Entropy (directed information) and Granger influence.
- Coupling between human prosody and feline posture/presence indices.

7.4 Presence Index

A minimal index combining humans (H_1, H_2), conversation M, and felines (F_1, F_2):

$$S_t = \alpha\, H_1 H_2 M + \beta\, F_1 F_2 - \delta\, F_1(1 - F_2). \tag{7.1}$$

This index conditions the influence models (VAR/TE) during sessions.

Chapter 8

Implementation Blueprint

8.1 Sensing and Tooling

- Video of humans/felines at fixed frame rate; audio at a common sample rate.
- Keypoints for felines (ears, head pose) and humans (face, gaze); prosody features.
- Time synchronization via monotonic clocks and per-frame timestamps.

8.2 Data Model

Store raw streams; derived features; symbols z_t; inferred states $\hat{s}_t$; presence vector (H_1, H_2, F_1, F_2, M); and event logs of cycles.

8.3 Pipeline

1. Preprocess $\rightarrow$ features ϕ.
2. Symbolize via σ to yield z_t.
3. Infer latent states $\hat{s}_t$ (e.g., HMM or Bayesian filters).
4. Estimate TE/VAR; compute loop gains; summarize IRA metrics.

Chapter 9

Applications

9.1 AI Architecture

Recursive AI with dynamic symbolic↔sensory loops, self-attention, and policy meta-updates112†source.

9.2 Cognitive/Therapeutic Use

Map distortions to branches; behaviors to leaves; use equilibria (R_1, R_2) as targets for progress111†source.

9.3 Ethology and Human–Animal Interaction

Model cat–human coupling; estimate how presence and prosody alter trust indices (T_{Tenchi}).

9.4 Creative Systems

Trigger artifacts (songs, psalms, MTG cards, diagrams) when salience+reward exceeds threshold; this is a formal generator in the state-space model113†source.

Chapter 10

Validation, Limitations, and Ethics

10.1 Validation

Hold-out sessions; shuffled-time controls (TE should drop to near zero); inter-annotator reliability for dialogue coding.

10.2 Limitations

Quantum↔cognition links are speculative; symbolic encodings can distort analog nuance; interpretability remains a challenge112†source.

10.3 Ethics

Non-invasive recording; animal welfare; consent; secure storage and anonymization for any public sharing.

Chapter 11

Conclusion

Recursive feedback loops unify tree logic, strange loops, quantum-inspired inference, feline–human interaction, and staged workflows. The framework is implementable without sacrificing analog richness, while remaining readable via externalized visual logic.

Appendix A

Notation and Equations (Collected)

From *Theory 2*

Key stage and equilibrium expressions: A, A_{31}^{t}, R_1, R_2.

From *Equation of the Mind*

Global state, attention/softmax, coupling, special modules, and artifact generator.

From *Strange Loops AI Paper*

Cross-disciplinary synthesis; quantum-inspired recursion; literature anchors.

From *Thought Tree Method*

Trunk/branch/leaf workflow and disciplined pruning.

Appendix B

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